

Pharmacoeconomics: Who Paid for Pakistan's Medicine Deregulation?

Medicine Prices, Manufacturer Margins and Patient Costs after the 2024 Reform

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KEY NUMBERS

32%

Overall rise in medicine prices since deregulation, as reported by the Health Ministry and DRAP to the Senate (Nov 2025)

69%

Share of the growth in pharmacy medicine sales that came from higher prices, not more packs sold (year to Mar 2025)

41%

Gross profit margin of listed drug manufacturers in 2025, up from 26% in 2023

Rs 1,904

Extra a patient pays each month when a doctor prescribes brands, not generics, for a common diabetes and heart regimen

Sources: Dawn, reporting the Ministry of National Health Services and DRAP; IQVIA data reported by Profit; Toplevel Securities; Ul Haq et al. (2026). Calculations by ARS.

Key Findings

1. Medicine prices rose sharply after deregulation. The Health Ministry and the Drug Regulatory Authority of Pakistan (DRAP) put the overall increase at 32%, and more than half of the 771 deregulated brands surveyed by DRAP were selling above their earlier price.
2. Pharmacy sales grew mainly because prices rose. Higher prices explained 69% of the growth in sales in the year to March 2025, while the number of packs sold rose by only 3.6%.
3. Prices rose most where competition was weakest. Original brands of diabetes tablets rose 47%, against 15% for the cheapest generics; insulins freed from price control rose 37%, against 22% for insulins still controlled.
4. Manufacturers' margins rose well beyond their earlier level. The gross margin of listed drug companies rose from 26% of sales in 2023 to 41% in 2025, while the rupee was stable and imported raw materials became cheaper.
5. Patients bore the cost. A common regimen of six medicines for diabetes, blood pressure and cholesterol costs a patient Rs 1,904 more each month when the doctor prescribes original brands rather than the cheapest generics.

About the Data

This brief asks a simple question: who paid when, in February 2024, the government ended price control on medicines outside the National Essential Medicines List (NEML)? Until then, DRAP fixed the maximum retail price of every registered medicine. Since then, manufacturers have set the prices of medicines outside the list themselves, while the prices of listed medicines remain controlled.^{1,2}

The analysis draws on six sources. Consumer prices come from the national Consumer Price Index (CPI) of the Pakistan Bureau of Statistics (PBS), available monthly from July 2022 to August 2026.³ Pharmacy sales come from IQVIA, a firm that audits retail pharmacy sales, as reported in the financial press.^{4,5} Changes in the prices of individual brands come from a DRAP survey of 771 deregulated brands, presented to the Senate Standing Committee on National Health Services in May 2026.⁶ Prices paid by patients come from three peer reviewed surveys of private pharmacies that follow the standard method of the World Health Organization and Health Action International (WHO/HAI).^{2,7,8,9} Manufacturers' results come from Toplevel Securities' compilation

of the published accounts of listed pharmaceutical companies.^{10,11,12} Health spending comes from the WHO Global Health Expenditure Database and the PBS National Health Accounts.^{13,14} Table 1 summarises how the pricing rules changed.^{15,16}

Table 1. Changes to Pakistan’s medicine pricing rules, 2018 to 2026

Date	Measure	What it did
2018	Drug Pricing Policy 2018	Government fixes the maximum retail price of every registered medicine. New medicines priced against the average in India and Bangladesh. Annual increases limited to 70% of inflation (maximum 7%) for essential medicines and 100% of inflation (maximum 10%) for others. Pharmacy margin set at 15% of the retail price. ¹⁵
May 2023	Single price increase	Prices allowed to rise by up to 14% for essential medicines and up to 20% for other medicines, after the fall in the rupee. ^{10,17}
Jun 2023	Wider definition of essential	Every strength of a medicine treated as essential if any strength appears on the WHO Model List. ¹⁸
Oct 2023	S.R.O. 1423(I)/2023	WHO Model List of Essential Medicines (2023) adopted as Pakistan’s essential list for pricing purposes. ¹⁶
Feb 2024	Federal cabinet decision (caretaker government)	Medicines not on the essential list freed from price control; manufacturers may set and revise their prices without prior approval. ¹²
Oct 2025	New national essential list	Model List of National Essential Medicines 2025 notified. ¹⁹
Mar 2026	Government review	Deregulation retained, with closer price monitoring promised. ^{20,21}
Sep 2026	Federal directive	Revision of the hardship policy and of DRAP’s Policy Board ordered; the industry opposes the changes. ²²

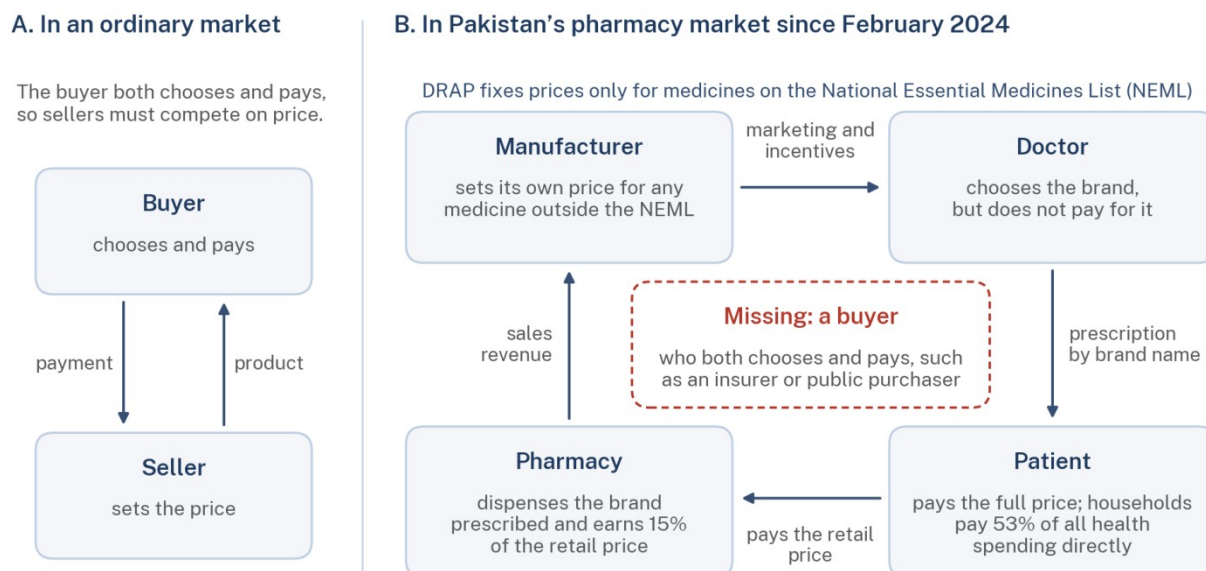
Note: Sources as cited in the table.

Three terms recur. An original brand is the product of the pharmaceutical company that first developed a medicine; a generic is a copy of the same medicine made by another company, and in Pakistan generics are also sold under brand names. Affordability is measured in days’ wages: the number of days a low paid government worker must work to buy one month’s supply. Gross margin is the share of sales revenue that remains after the cost of making the product.

Why Deregulation Did Not Create Price Competition

The case for deregulation rests on a simple idea: when sellers are free to set prices, buyers who compare prices will force those prices down. The idea assumes that the person who chooses a product is also the person who pays. For most medicines in Pakistan, that assumption does not hold (Figure 1). The doctor chooses the medicine but does not pay. The patient pays but seldom chooses, and usually cannot judge whether a cheaper product would work as well. No insurer or government purchaser stands between the two. In 2023, 52.9% of all health spending was paid directly by households rather than by the government or insurance companies,¹³ and the national health insurance programme, Sehat Sahulat, covers hospital admission, not the medicines that people with long term illness buy from pharmacies.⁹ The pharmacy, whose margin is set at 15% of the printed retail price, earns more when it sells the more expensive brand.¹⁵

Figure 1. In Pakistan's pharmacy market, the doctor chooses the medicine but the patient pays



Note: NEML is the National Essential Medicines List. The household share is direct (out of pocket) spending as a share of current health expenditure in 2023. The pharmacy margin is the retailer's discount of 15% on the maximum retail price. Sources: WHO Global Health Expenditure Database;¹³ DRAP, Drug Pricing Policy 2018.¹⁵ ARS analysis.

Economists have long recognised this problem. Kenneth Arrow showed in 1963 that patients must rely on doctors to choose treatment on their behalf, because they cannot judge its quality in advance.²³

Later research found that the doctor's choice responds to the doctor's own incentives as well as to the patient's needs.²⁴ In Japan, for example, doctors who also dispensed medicines prescribed more of the products that earned them a higher margin, and were slower to switch patients to cheaper generics.^{25,26}

Pakistan's market has many sellers but little competition on price. More than 76,000 brands of about 1,600 medicines and combinations are registered,²⁷ and companies compete mainly for doctors' prescriptions rather than for patients' purchases. In a randomised trial among private doctors in Karachi, 41.9% of those in the comparison group agreed to accept incentives from pharmaceutical companies in return for meeting prescription targets.²⁸ Interview studies describe such arrangements as common practice.^{29,30} This brief therefore tests a clear proposition: where the person who chooses a medicine does not pay for it, removing price control is likely to raise the prices that patients pay.

Medicine and Medical Care Prices Rose Faster than Other Prices

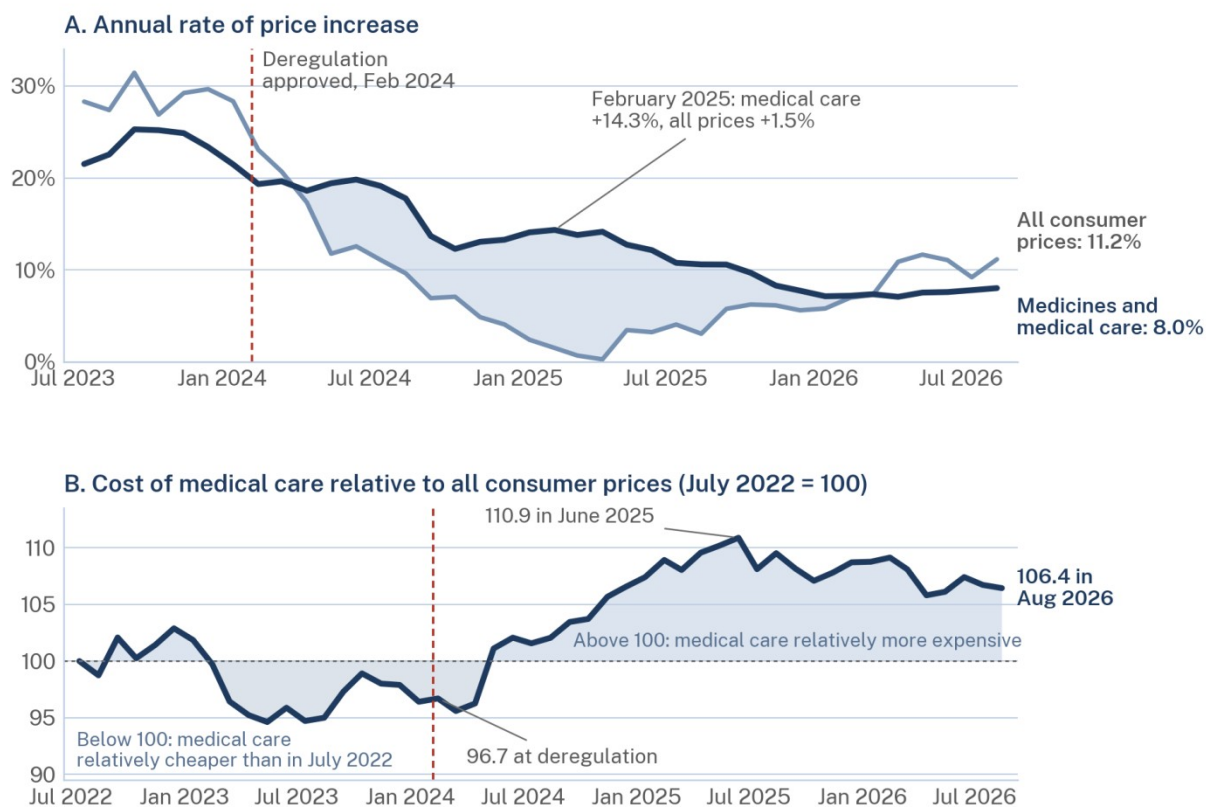
The CPI does not publish a monthly index for medicines alone. It combines medicines with doctors' fees, hospital charges and medical tests in a single health group, which carries a weight of 2.79% in the national index.³ This brief refers to that group as medical care prices. It is broader than a pure medicine price index, but it is the only official monthly series available.

Before deregulation, medical care prices rose more slowly than prices in general. Between July 2022 and February 2024 they rose 37.1%, while the overall CPI rose 41.8%. Over the same period the rupee fell from an average of 163 to the US dollar in 2021 to 280 in 2023.³¹ This raised the cost of imported raw materials, while controlled prices could rise only by a fixed formula. Manufacturers' complaints that many controlled prices had become unprofitable were, on this evidence, justified.

After deregulation the pattern reversed (Figure 2). Medical care prices rose faster than overall prices in 24 of the 30 months from March 2024 to August 2026. In February 2025, for example, they were 14.3% higher than a year earlier, while overall consumer prices had risen by only 1.5%. Between February 2024 and February 2026, medical care prices rose 22.6%, compared with 8.6% for all consumer prices.

Only part of this increase can be explained as a correction for the earlier squeeze on controlled prices. Relative to other consumer prices, medical care was 3.3% cheaper in February 2024 than in July 2022. By June 2025 it was 10.9% more expensive. The first 3.3 percentage points restored the balance that existed before the currency crisis. The remaining 10.9 points took medical care prices well beyond that balance, and they rose while the rupee was stable at 279 to 281 to the US dollar.^{31,32} The difference narrowed in 2026, when overall inflation rose above 11%, but in August 2026 medical care remained 6.4% more expensive, relative to other prices, than in July 2022.

Figure 2. After deregulation, the cost of medicines and medical care rose faster than consumer prices in general



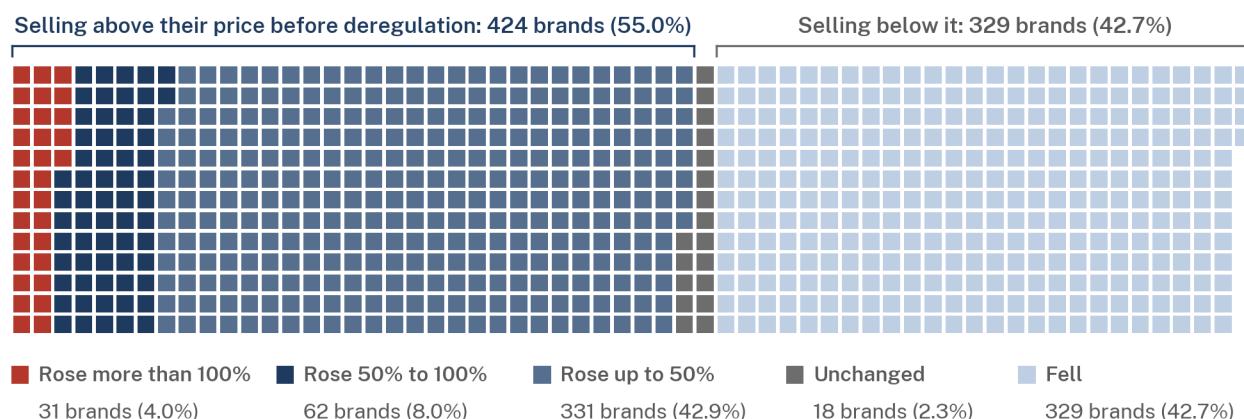
Note: National CPI, base 2015-16. Medical care is the CPI health group (weight 2.79%), which covers medicines, doctors' fees, hospital charges and medical tests. Panel B divides the medical care index by the index for all items, with both set to 100 in July 2022; a value above 100 means medical care has become more expensive relative to other goods and services. Source: PBS CPI data.³ ARS calculations.

Most Deregulated Brands Became More Expensive

DRAP's own survey provides the most direct evidence. In May 2026 DRAP reported to the Senate committee on 771 brands representing the 500 best selling medicines outside the essential list.⁶ Of these, 424 (55.0%) were selling above their price before deregulation, 329 (42.7%) below it and 18 at the same price (Figure 3). Of the brands that became more expensive, 331 rose by up to 50%, 62 by between 50% and 100%, and 31 by more than 100%.³³ About one brand in eight therefore rose in price by more than half. DRAP presented the survey as evidence that medicines were again available, since every brand surveyed was on sale; the committee asked the Competition Commission of Pakistan to review the increases.^{6,33}

The survey has two limitations. It does not weight brands by sales, so it cannot show whether patients mainly bought the brands that became more expensive or those that became cheaper. It also compares current prices with the maximum prices previously notified, which may exceed what pharmacies actually charged. The Health Ministry and DRAP gave the same committee a broader estimate in November 2025: an overall increase of 32% in medicine prices since deregulation.³⁴

Figure 3. More than half of 771 deregulated brands were selling above their price before deregulation



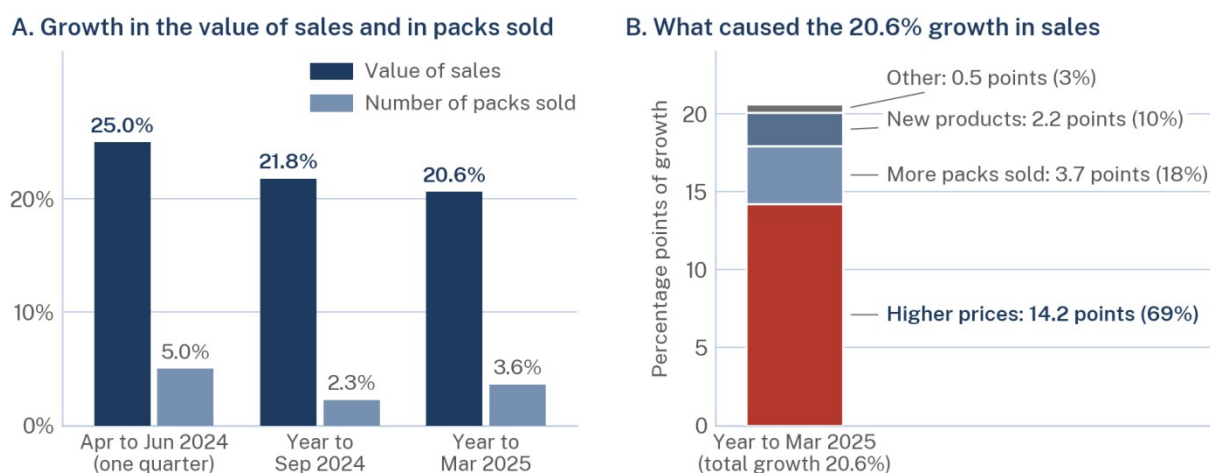
Each square represents one brand. Price changes compare the current price with the price before deregulation.

Note: DRAP survey of 771 brands representing the 500 most widely sold medicines outside the essential list, presented to the Senate Standing Committee on National Health Services, Regulations and Coordination in May 2026. The counts in each price band are as reported in press accounts; DRAP has not published the presentation. Sources: Business Recorder;⁶ 24 News HD.³³

Sales Grew Because Prices Rose, Not Because More Medicines Were Sold

Pharmacy sales data point in the same direction. In the year to March 2025, pharmacy sales of medicines reached Rs 1.049 trillion, 20.6% more than a year earlier, but the number of packs sold rose by only 3.6%.⁴ IQVIA attributed 69% of the growth to higher prices, 18% to more packs sold and 10% to new products (Figure 4). The same pattern appeared from the first months of deregulation: in April to June 2024, sales were 25% higher than a year earlier, of which 20 percentage points came from higher prices.^{5,35} Measured in US dollars, the market grew by only about 4% a year over the four years to March 2025.⁴

Figure 4. Higher prices, not higher volumes, drove the growth of pharmacy sales



Note: Panel A compares growth in the value of pharmacy sales with growth in the number of packs sold; the first period is a single quarter compared with the same quarter a year earlier. Panel B divides the 20.6% growth in the year to March 2025 into the contributions reported by IQVIA; “other” is the part not allocated. Sources: IQVIA data reported by Profit^{4,5} and Business Recorder.³⁵ ARS calculations.

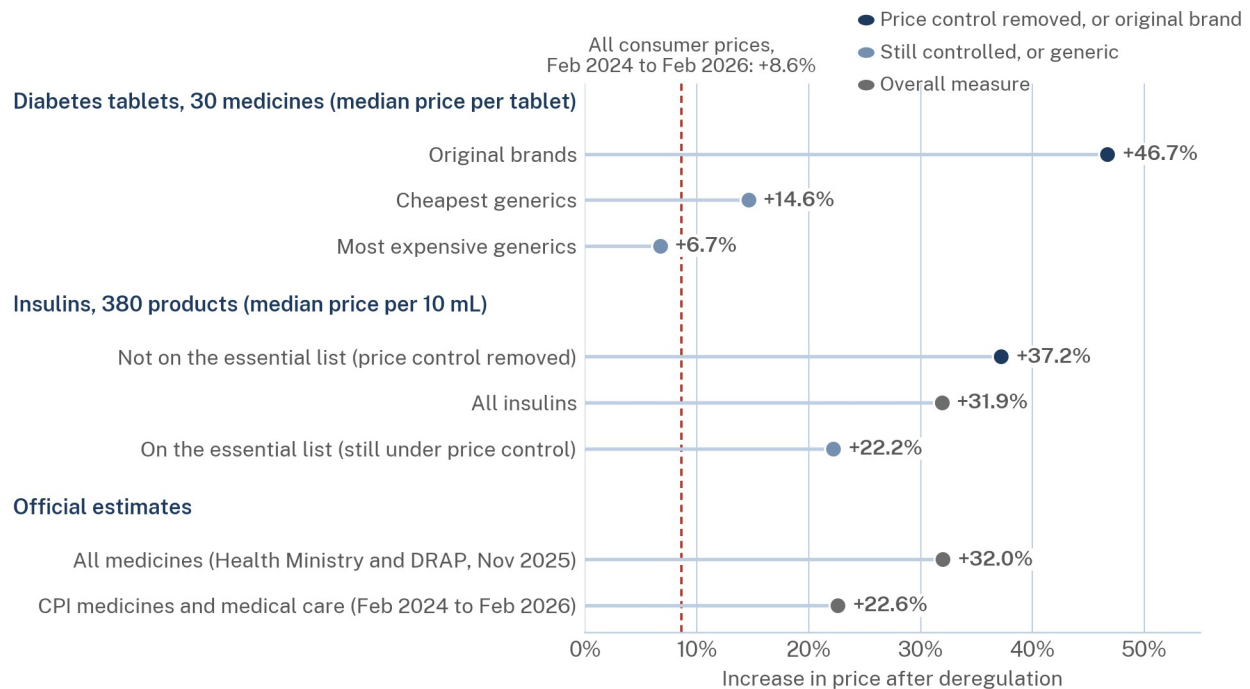
This matters because patients cannot easily cut back on prescribed treatment for long term conditions. When prices rise, most continue to buy, so a higher price becomes higher spending rather than lower consumption. Growth of 2% to 4% a year in the number of packs sold, in a population that is growing and ageing, gives no sign that higher prices brought many more patients into treatment.

Prices Rose Most Where Competition Was Weakest

Two peer reviewed studies measured prices in 30 private pharmacies in six cities before deregulation and again in June and July 2024.^{2,8} Both compared medicines freed from price control with medicines that remained controlled, a method that separates the effect of deregulation from general inflation. For 30 diabetes tablets, the median price of original brands rose 46.7%, that of the cheapest generics 14.6% and that of the most expensive generics 6.7% (Figure 5). For 380 insulin products, prices of insulin outside the essential list rose 37.2%, against 22.2% for insulin that remained on the controlled list. The cost of a month's supply of the median insulin rose from 1.90 to 2.51 days' wages. Both studies found that a statistically significant part of the increase was caused by deregulation itself.

Competition explains the pattern. In the diabetes study, each controlled medicine had on average 37 competing registered products, while each deregulated medicine had only 9.⁸ Deregulation therefore removed price control where competition was weakest. Comparing versions of the same medicine, original brands became more expensive faster than generics. Their buyers are the least sensitive to price, because the choice follows the doctor's preference rather than the price on the shelf. Research in the United States has documented the same effect: when cheaper generics enter the market, original brands can raise their prices, because they keep the patients whose doctors continue to prescribe the brand.³⁶ The insulin market is also highly concentrated, with three multinational companies supplying 85% of the products surveyed.² Evidence from India points to a further risk: when only some forms of a medicine are price controlled, companies can shift sales toward the forms that are not.³⁷

Figure 5. Price increases were largest for original brands and for medicines freed from price control



Note: The two studies compare median prices in 30 private pharmacies in six cities before deregulation and in June and July 2024. The official estimates cover longer periods and are shown for comparison. Sources: Saeed et al. (2025);⁸ Saeed et al. (2026);² Dawn;³⁴ PBS.³ ARS calculations.

The line between controlled and uncontrolled medicines was itself imported. The essential list used to draw that line was the WHO Model List of Essential Medicines, adopted in October 2023 without adaptation to Pakistan’s needs.^{2,16} WHO, however, intends its model list as a starting point that each country should adapt to its own disease burden.³⁸ Pakistan has the highest rate of diabetes in the world, affecting 31.4% of adults,³⁹ and 34% of adults aged 30 to 79 have high blood pressure.⁴⁰ Yet in the diabetes study, 19 of the 30 medicines on sale were outside the list and therefore free of price control. A new national list was notified in October 2025;¹⁹ this brief could not yet assess whether the new list reflects Pakistan’s own disease burden.

The Doctor’s Choice of Brand Largely Determines the Patient’s Bill

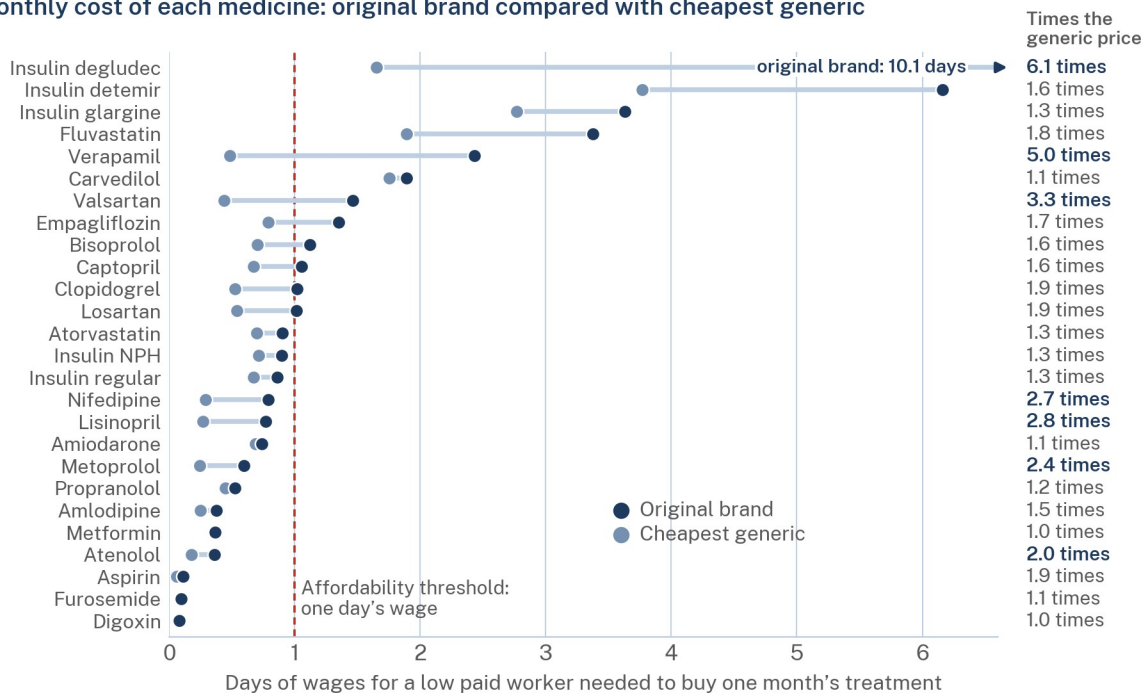
The most recent survey, conducted in October and November 2025 in six districts, compares the monthly cost of original brands and the cheapest generics for 26 medicines used to treat heart disease and diabetes.⁹ The median original brand cost 1.6 times as much as the cheapest generic, and for some medicines far more: 6.1 times for insulin degludec and 5.0 times for verapamil (Figure 6). WHO/HAI regard a month’s treatment as affordable if it costs no more than one day’s wages. By that standard, 12 of the 26 medicines were unaffordable as original brands and 5 as generics. International surveys have found similar gaps between brands and generics in many developing countries.⁴¹

Patients with diabetes and heart disease rarely take a single medicine. Consider a common combination for a patient with type 2 diabetes, high blood pressure and raised cholesterol:

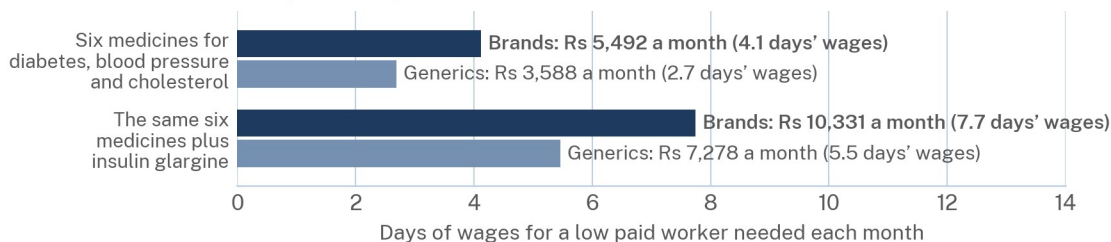
metformin, empagliflozin, amlodipine, losartan, atorvastatin and aspirin. At the survey's prices, it costs Rs 5,492 a month in original brands and Rs 3,588 in the cheapest generics. The difference of Rs 1,904 a month, or Rs 22,846 a year, equals about 17 days' wages for a low paid worker, and it depends almost entirely on the brand the doctor writes on the prescription. Adding insulin glargine raises the monthly cost to Rs 10,331 in brands and Rs 7,278 in generics. At the survey's reference wage of Rs 40,000 a month, the branded regimen with insulin takes a quarter of the income of a worker on a low wage.

Figure 6. Original brands cost much more than generics, and the difference adds up across a typical regimen

A. Monthly cost of each medicine: original brand compared with cheapest generic



B. The difference adds up when a patient takes several medicines



Note: Private pharmacies, October to the middle of November 2025. Days' wages use the survey's daily wage of Rs 1,333 for the lowest paid unskilled government worker. The source reports two regimens for captopril; one is shown. Panel B adds up the survey's median prices for the medicines listed; the regimen is an illustration, not clinical guidance. Source: Ul Haq et al. (2026), Table 2.⁹ ARS calculations.

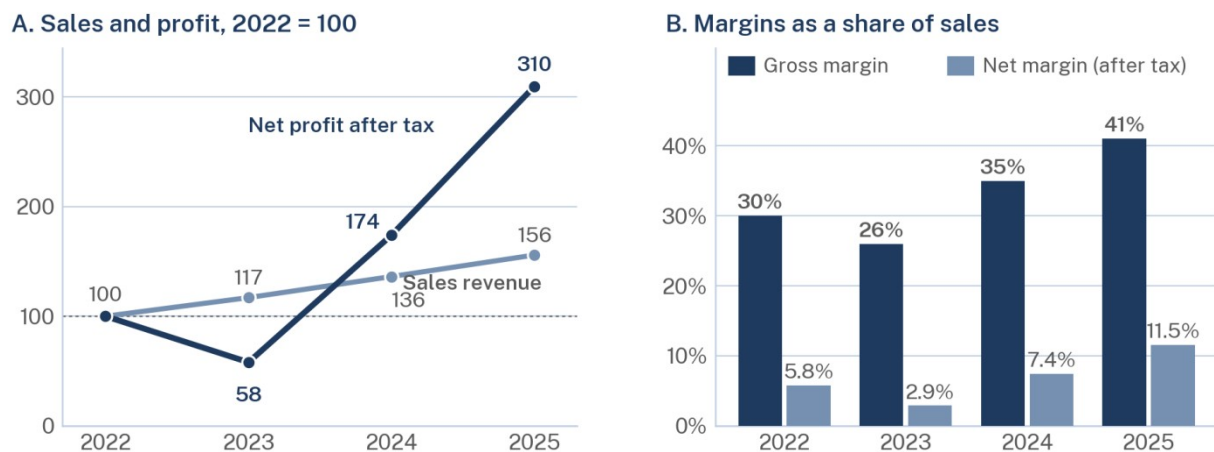
Manufacturers' Margins Rose Well Beyond Their Earlier Level

The financial results of listed drug manufacturers show who gained from the higher prices. According to Toplevel Securities, their gross margin fell from 30% of sales in 2022 to 26% in 2023, as the rupee fell and controlled prices lagged behind costs.¹⁰ It then rose to 35% in 2024 and 41% in 2025. Net profit after tax increased from Rs 7.9 billion in 2023 to Rs 42.2 billion in 2025, while sales grew by 33% (Figure 7).^{11,12} Had the 2023 margin remained unchanged, gross profit in 2025 would have been about Rs 95 billion; at the actual margin it was about Rs 150 billion. About 70% of the

increase in gross profit between 2023 and 2025 therefore came from higher margins, and about 30% from higher sales.

Rising costs do not explain the higher margins. The rupee averaged 280 to the US dollar in 2023 and 281 in 2025,³¹ and between January and October 2025, 53% of the active ingredients used by the industry became cheaper, by a median of 11%.⁴² Because Pakistan imports 80% to 90% of its pharmaceutical raw materials, a stable rupee and cheaper ingredients reduce costs across the industry.⁴³

Figure 7. Listed manufacturers' profits rose much faster than their sales



Note: Totals for the listed pharmaceutical companies covered by Topline Securities. Values for 2022 are derived from the growth rates reported for 2023, and the companies covered may differ slightly between annual reviews. Sources: Topline Securities, as reported by Business Recorder^{10,12} and ProPakistani.¹¹ ARS calculations.

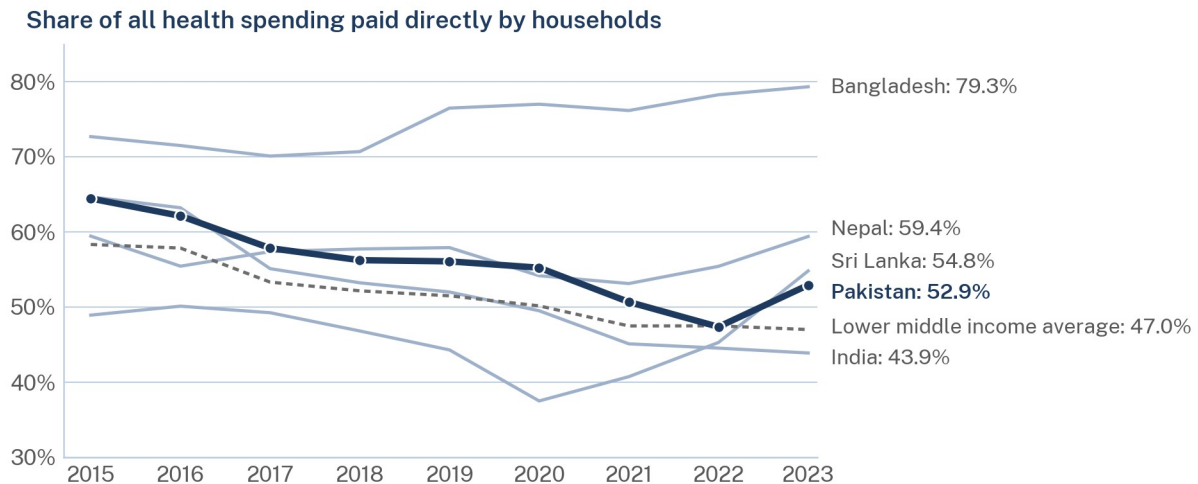
The industry's position deserves fair consideration, and part of it is supported by the evidence. The industry reports that its net profit margin was about 3% before 2024; the listed companies' accounts show 2.9% in 2023, a level that could not be sustained.²² Officials acknowledge that controlled prices which failed to keep pace with the rupee led manufacturers to cut or stop production, and that deregulation reduced shortages of routine medicines.²⁰ The Health Minister told the National Assembly that five major manufacturers, most of them multinationals, had left manufacturing in Pakistan over three years, although two of the companies disputed that account.⁴⁴ Exports rose 34% to USD 457 million in FY2025, the fastest growth in two decades, although they remain equal to about one eighth of domestic pharmacy sales.^{4,45} The question is therefore not whether margins needed to recover, but by how much and at whose expense. By 2025 the gross margin was 11 percentage points above its 2022 level, and patients paid for that increase.

Households Bore the Cost

In Pakistan, higher medicine prices fall mainly on households. Households paid 52.9% of all health spending directly in 2023, up from 47.4% in 2022 and above the average of 47.0% for lower

middle income countries (Figure 8).¹³ The National Health Accounts put this direct household spending at Rs 917 billion in 2021-22,¹⁴ and earlier research found that medicines account for more than half of the total.⁴⁶ Because the national health insurance scheme covers hospital care, patients pay almost the entire cost of the medicines for long term illness that they buy from pharmacies.

Figure 8. Pakistani households pay more than half of all health spending directly



Note: Direct (out of pocket) payments by households as a share of current health expenditure. Source: WHO Global Health Expenditure Database, via World Bank World Development Indicators.¹³

These costs affect both household budgets and health. A study using the national household survey found that when households spend more on medicines for blood pressure and diabetes, they spend less on food and education.⁴⁶ WHO estimates that of every 100 adults aged 30 to 79 with high blood pressure in Pakistan, 54 have been diagnosed, 25 are receiving treatment and only 8 have their blood pressure under control.⁴⁰ Cost alone does not explain this gap, but it is the factor most within reach of pricing policy. The patients who paid most for deregulation were those with long term conditions such as diabetes and heart disease, taking branded medicines, paying at the pharmacy without insurance and with little income to spare.

Shortages Persist among Essential Medicines under Price Control

Deregulation created two markets with opposite problems. Essential medicines, about 500 molecules and 40% of the market according to the industry, remain under a formula that allows prices to rise each year by 70% of the inflation rate; in FY2026 this permitted an increase of no more than 4.9%.²² The Drug Pricing Policy requires DRAP to decide applications for price relief on grounds of hardship within 120 days,¹⁵ yet the industry reports applications pending since February 2024. In July 2026 the Pakistan Chemists and Druggists Association reported 105 essential medicines, including oral morphine, streptokinase and several cancer medicines, as unavailable or in very short supply.⁴⁷ DRAP stated in March 2026 that no essential medicine had received a price increase.⁴⁸

The reform therefore gave pricing freedom where competition is weakest and patients are least able to resist higher prices, while leaving rigid controls where medicines are most essential and shortages most dangerous. Patients who cannot find a controlled medicine also pay, through more expensive substitutes, delayed treatment or purchases from informal sellers.

Policy Implications

The evidence supports four conclusions, which Table 2 matches with policy options. Studies of pricing policy in low and middle income countries reach a similar view: price regulation works only when it is combined with price monitoring, policies that promote generics and a purchaser able to act on the information.⁴⁹

First, deregulation changed the prices of medicines without changing the way the market works, and in this market higher prices were the predictable result. A return to blanket price control is not the answer: it caused the shortages that led to deregulation, and strict price control can itself weaken competition among generics.⁵⁰ A more effective response is to strengthen the buyer's side of the market. This means requiring doctors to prescribe by generic name, allowing pharmacists to dispense a cheaper generic of assured quality, and publishing a reference price for each group of medicines that treat the same condition. WHO strongly recommends policies that promote quality assured generics,⁵¹ and such policies reduced generic prices by more than 80% in several lower income European countries.²⁷ Doctors and patients will trust generic substitution only if DRAP enforces bioequivalence testing, which shows that each generic works as well as the original brand.²⁷

Second, the essential list should reflect Pakistan's disease burden rather than a model list designed for global use. A list built around the country's leading diseases would bring the combinations and newer medicines widely prescribed for diabetes and heart disease into the controlled or reference priced group, while leaving truly discretionary products to the market. India offers one approach: it sets the ceiling price of each controlled medicine at the average price of all brands with at least 1% of sales, and limits annual increases for other medicines to 10%.⁵²

Third, the controlled group needs a pricing rule that keeps medicines on the shelves. A formula that allows only 70% of inflation guarantees losses in years of high inflation, and a hardship process that leaves applications undecided for two years turns price control into shortage. Automatic adjustment for changes in input costs, together with hardship decisions within the 120 days the policy already requires, would protect supply without reversing deregulation.

Fourth, medicines for long term illness need a payer. Pooled purchasing or insurance cover for a short list of essential outpatient medicines would create a buyer with the information and the incentive to negotiate lower prices.³⁸ For doctors, the implication is direct: in this market the

prescription largely determines the price a patient pays, and prescribing by generic name is the most effective cost control available to patients today.

Table 2. Policy options to reduce what patients pay for medicines, and the problem each option addresses

Problem	Policy option	Evidence or precedent
Doctors choose medicines but patients pay	Prescribing by generic name, with pharmacists allowed to dispense a generic of assured quality	Such policies cut generic statin prices by 83% to 87% in Lithuania ²⁷
Pharmacies earn more when they sell more expensive brands	A fixed dispensing fee in place of a 15% margin on the retail price	WHO recommends regulating markups in the supply chain ⁵¹
The essential list was adopted from WHO without national adaptation	A national essential list based on Pakistan's disease burden and reviewed regularly	WHO designs its model list for adaptation by each country ³⁸
Deregulated medicines face little competition	A published reference price for each group of medicines that treat the same condition	India sets ceilings at the average price of brands with at least 1% of sales ⁵²
Controlled prices fall behind costs, causing shortages	Automatic adjustment for input costs and firm deadlines for hardship applications	The 2018 policy already requires hardship decisions within 120 days ¹⁵
No one buys outpatient medicines on patients' behalf	Pooled purchasing or insurance cover for a short list of medicines for chronic illness	Recommended by the Lancet Commission on essential medicines ³⁸

Note: Policy options draw on the WHO guideline on country pharmaceutical pricing policies and the evidence cited in this brief. ARS analysis.

Data Notes

CPI figures are from the PBS national index, base 2015-16; annual changes and the relative price index were calculated for this brief. Because the health group includes doctors' fees and hospital charges as well as medicines, it may overstate or understate the change in medicine prices alone. The DRAP survey is not weighted by sales and compares market prices with previously notified maximum prices; the shares DRAP reported (55.06%, 42.67% and 2.27%) differ slightly from shares calculated from the reported counts, and this brief uses the counts. IQVIA does not publish its Pakistan audit, so its figures are taken from press reports that cite the audit. The 2024 pharmacy surveys used a monthly wage of Rs 32,000 and the 2025 survey a daily wage of Rs 1,333, so days' wages can be compared within each survey but not between surveys. The regimen of six medicines adds up the survey's median prices at the doses the survey used; it illustrates cumulative cost and is not a treatment recommendation. Totals for listed companies depend on the companies covered in each annual review and do not represent the whole industry.

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